

Samir Khanal

Dear Hiring Team,

I'm applying for the AI/ML Intern position because the work described in the role closely matches the direction I've been building toward in data science, machine learning, NLP, and Generative AI.

Recently, I identified one gap in my experience: I understood RAG architecture and its underlying concepts, but I wanted stronger hands-on experience building the complete system myself. I addressed that by building a Hybrid SQL + RAG Knowledge Assistant. The system combines text-to-SQL for structured data, semantic retrieval using ChromaDB and Sentence Transformer embeddings for documents, a semantic routing layer, Gemini API integration, and a FastAPI backend. I also implemented SQL validation so LLM-generated queries are restricted to safe read-only operations.

The project reflects how I approach unfamiliar technical problems. Rather than stopping at understanding a concept, I try to turn it into something I can build, test, and explain. At CloudTech, I took ownership of documentation for a system I was still learning and developed it into the team's primary sprint reference. At Nepal Telecom, I worked on the research and design side of a RAG-based chatbot and developed a deeper understanding of document retrieval, embeddings, and LLM-based workflows. These experiences have taught me that I don't need to know everything on day one to contribute; I need to be willing to understand the problem, learn what is missing, and follow it through to a working result.

I'd welcome the opportunity to discuss the project and how I could bring that same approach to the AI/ML team. Thank you for your time.

Samir Khanal